

SKILLS

Game & XR: Unity (C#), Vuforia AR SDK, WebXR, Three.js, ML-Agents
Backend: FastAPI, Flask, REST API, WebSocket, RabbitMQ
DevOps: Docker, Jenkins, GitHub Actions
Mobile: React Native, Android (Java)

AI & ML: LLaMA-3.1, LangChain, Groq, MediaPipe FaceMesh, LSTM, scikit-learn
Standards: xAPI, SCORM, LMS integration (Moodle, Canvas, Blackboard)
Languages / DB: C#, Python, Java, JS · PostgreSQL, MySQL
Languages: Arabic (Native) · French (B2) · English (C1)

PROJECTS

[PFE] Adaptive Serious Game — Generative AI & Emotion Detection Platform 2026–Present

Unity · C# · LLaMA-3.1 · LangChain · Groq · FastAPI · MediaPipe FaceMesh · PostgreSQL · xAPI · WebSocket

- Generative AI (LLaMA-3.1/LangChain) dynamically creates questions, scenarios and adaptive feedback in real time based on learner emotional state (MediaPipe FaceMesh) and performance data
- Intelligent recommendation engine: adaptive difficulty, player profile tracking, personalized learning path generation — fully integrated in Unity 2D ↔ FastAPI real-time loop
- Full xAPI analytics pipeline, 9-table PostgreSQL schema, REST/WebSocket bridge — compatible with all major LMS platforms (Moodle, Canvas, Blackboard, SCORM)

VR/AR Educational Platform 2023–2024

Unity · Vuforia AR SDK · C# · VR/AR · 3D Environments

- Multi-scene VR immersive environments for STEM education (geometry, anatomy, biology); marker-based and markerless AR apps deployed and tested on mobile
- Interactive AR lab simulations: students manipulate 3D objects in their real environment via Vuforia

Directeur de Jeu — LLM Game Master (Beta) 2024

Unity · C# · LLM · Python · FastAPI

- AI-powered Game Master: LLM generates dynamic narrative content, adaptive events and branching dialogue in real time; designed the Unity ↔ FastAPI communication layer

Urban Simulation & AI Platform 2023

Python · Flask · ML · scikit-learn · Data Visualization

- ML-powered digital twin with Flask REST API, real-time environmental data ingestion and monitoring dashboards

Pixel Zombie Game 2023

Java · 2D Game Development

- 3-level zombie game with custom game loop, sprite rendering, collision detection and enemy AI state machine

EXPERIENCE

IT Teacher — 4 months Lycée Al Massar, Marrakech · 2025

- Taught IT fundamentals, programming basics and software services to high school students

IT Support Technician — 3 months Collège ELIA ABOU MADI, Marrakech · 2024

- Configured & deployed laptops; technical helpdesk support across departments

EDUCATION & CERTIFICATIONS

Master's — Engineering and Emerging Technologies, UCA-ENS Marrakech 2024–Present

Bachelor's — Computer Science, UCA-ENS Marrakech 2021–2024

MLIAEdu Certifications: NLP in Python · React Front-End · React Native · Java Advanced · Hibernate & JPA (Mar. 2026)